

Speed and Correctness Guarantees for Programmable Enthalpy-Neutral DNA Reactions[†]

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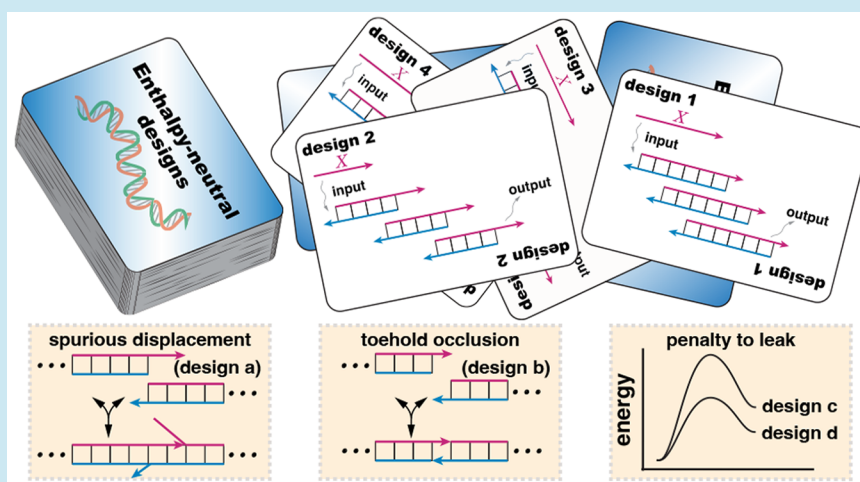
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ABSTRACT: Molecular control circuits embedded within chemical systems to direct molecular events have transformative applications in synthetic biology, medicine, and other fields. However, it is challenging to understand the collective behavior of components due to the combinatorial complexity of possible interactions. Some of the largest engineered molecular systems to date have been constructed using DNA strand displacement reactions, in which signals can be propagated without a net change in base pairs (enthalpy neutral). This flexible and programmable component has been used for constructing molecular logic circuits, smart structures and devices, for systems with complex autonomously generated dynamics, and for diagnostics. Limiting their utility, however, strand displacement systems are susceptible to the spurious release of output in the absence of the proper combination of inputs (leak), as well as reversible unproductive binding (toehold occlusion) and spurious displacement that slow down desired kinetics. We systematize the properties of the simplest enthalpy-neutral strand displacement cascades (logically linear topology), and develop a taxonomy for the desired and undesired properties affecting speed and correctness, and trade-offs between them based on a few fundamental parameters. We also show that enthalpy-neutral linear cascades can be engineered with stronger thermodynamic guarantees to leak than non-enthalpy-neutral designs. We confirm our theoretical analysis with laboratory experiments comparing the properties of different design parameters. Our method of tackling the combinatorial complexity using mathematical proofs can guide the engineering of robust and efficient molecular algorithms.

KEYWORDS: molecular programming, DNA nanotechnology, strand displacement, leak

INTRODUCTION

Nature evolved sophisticated molecular systems with diverse functional behaviors, ranging from simple chemical switches to vast gene regulatory networks. Inspired by nature, artificial molecular systems have shown significant promise in approaching this level of complexity. Guided by ideas from computer science and electrical engineering, molecular systems constructed from first-principles are finding applications in chemistry, material sciences, and medicine. Cascades of DNA strand displacement reactions have been shown to be a powerful mechanism for engineering molecular information

processing and dynamics.¹ In strand displacement cascades, a DNA strand displaces another strand from a multistranded prehybridized complex, and the displaced strand can in turn act as a displacing strand for downstream reactions.² Strand

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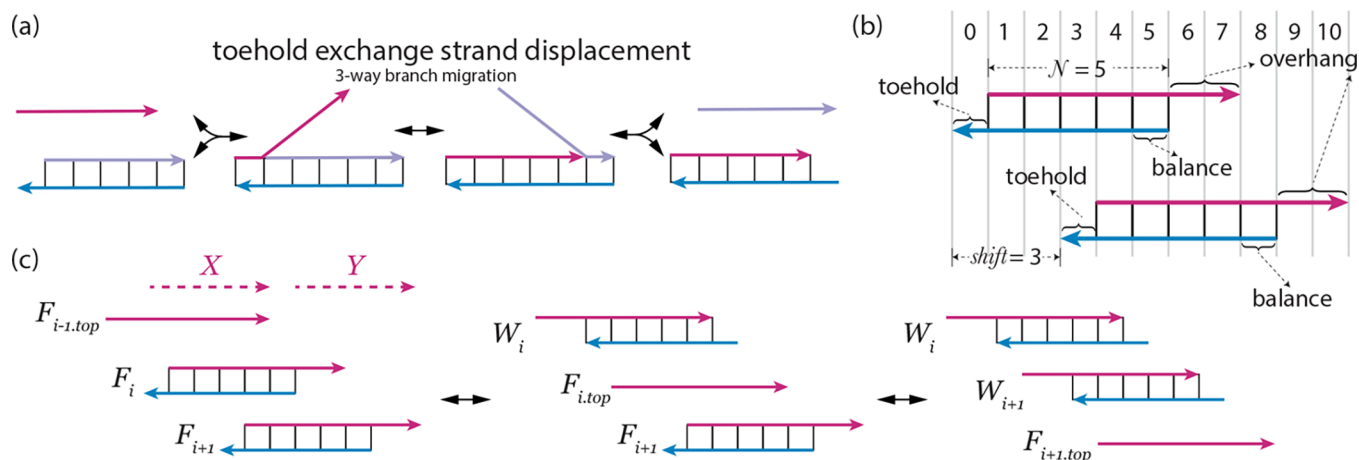


Figure 1. (a) The fundamental reaction step we consider. The toehold exchange reaction is shown in isolation and may occur in more complex scenarios between parts of larger complexes. (b) The conventions of the *EN* design. (c) Desired reaction pathway of a translator cascade ($X \rightarrow Y$). After 2 strand displacement steps, the signal strand with output region X (input to fuel F_i) is translated to the signal strand with output region Y (input to downstream fuel F_{i+2}).

displacement cascades have yielded some of the largest biochemical systems designed from scratch.³

Many desired and undesired properties of strand displacement systems are expressed and understood in terms of arguments about enthalpy and entropy, where the first is shorthand for the number of *bonds* and the second for the number of *separate complexes*, where “complex” refers to species of one or more strands. For example, an important obstacle to scaling up strand displacement systems is *leak*, which occurs when undesired reactions get spontaneously triggered in the absence of the correct input. A common approach to lowering leak involves adding one to three nucleotides (a *clamp*) at the end of helices which must break in the process of leak, leading to an enthalpic barrier.^{3–7} Since leak results from a spurious interaction between different complexes, lowering their concentration increases the entropic barrier to leak. In typical strand displacement systems, leak can result from a spurious interaction between just two complexes. In a recent leak reduction method (“*NLD* design”),^{8,9} the entropic penalty to leak can be programmed to be higher. Physically separating species that may react spuriously also increases the entropic barrier to leak.¹⁰ Other undesired properties include slowing of the desired reaction pathways due to the temporary sequestration of complexes in unreactive states (“toehold occlusion” and “spurious displacement”, see below). Almost universally, complexes have parts that are complementary by design, and thus they are driven to bind together by enthalpy, making some amount of sequestration unavoidable.

The driving force for the desired strand displacement reactions is often enthalpic—the formation of 5–7 new base pairs called a toehold. However, a type of strand displacement called *toehold-exchange* can preserve the overall number of base pairs.¹¹ Such enthalpy-neutral strand displacement cascades have played an important role in *de novo* engineering of molecular systems. Driven forward solely by the entropy of forming additional separate complexes, molecular signals can get exponentially amplified.¹² In the form of “see-saw gates”, enthalpy-neutral strand displacement was a key module in chemical circuits for performing Boolean logic computation³ and neural network computation.^{13,14} Molecular robots that can move along a track and sort cargo¹⁵ have also relied on

enthalpy-neutral strand displacement modules. Since mismatches between the invading strand and its target can significantly affect the equilibrium of enthalpy-neutral displacement, single-nucleotide mismatches can be detected with high specificity,¹⁶ giving rise to potential medical applications.^{17,18} Besides these experimental implementations, theoretical work on implementing reversible reactions with DNA strand displacement extensively uses enthalpy-neutral reactions.^{19,20}

Even the simplest kind of enthalpy-neutral strand displacement cascades—a cascade of translators, which are logically equivalent to repeater gates—can already exhibit complex and useful behavior. Translators can be composed to compute the logic OR of multiple distinct inputs translated to a common output.⁹ A catalytic system can also be constructed if a translator chain’s output is the same as its input. Indeed, the “see-saw” networks discussed above combine enthalpy-neutral translators with single-stranded fuels and threshold gates to achieve their powerful functionality.

Here we demonstrate an approach to systematically understand desired and undesired properties of strand displacement cascades based on combinatorial arguments about structure at the level of domains (DNA bases that act as a unit), enthalpy and entropy. We focus on cascades of enthalpy-neutral (*EN*) translators with logically linear topology (no fan-in or fan-out), and show that the entire design space can be captured by adjusting two parameters related to the number of double-stranded domains in the prehybridized complexes and how many domains consecutive complexes in the cascade share. Each of the variety of designs achieved has unique properties. Some parameter choices lead to toehold occlusion^{3,7} (blocking the toehold for a desired reaction), and spurious strand displacement (partial displacement of a strand on a complex by a spurious invader), both of which are likely to hinder the kinetics of the intended reaction pathway. We further develop combinatorial arguments about entropy and enthalpy that lead to a general analysis of leak in *EN* cascades. By counting the number of bonds and separate complexes, our theoretical arguments show that the thermodynamic penalty to leak via its enthalpic and entropic components can be made arbitrarily large with a proper choice of design parameters. Compared to the *NLD* design, enthalpy-neutral cascades are

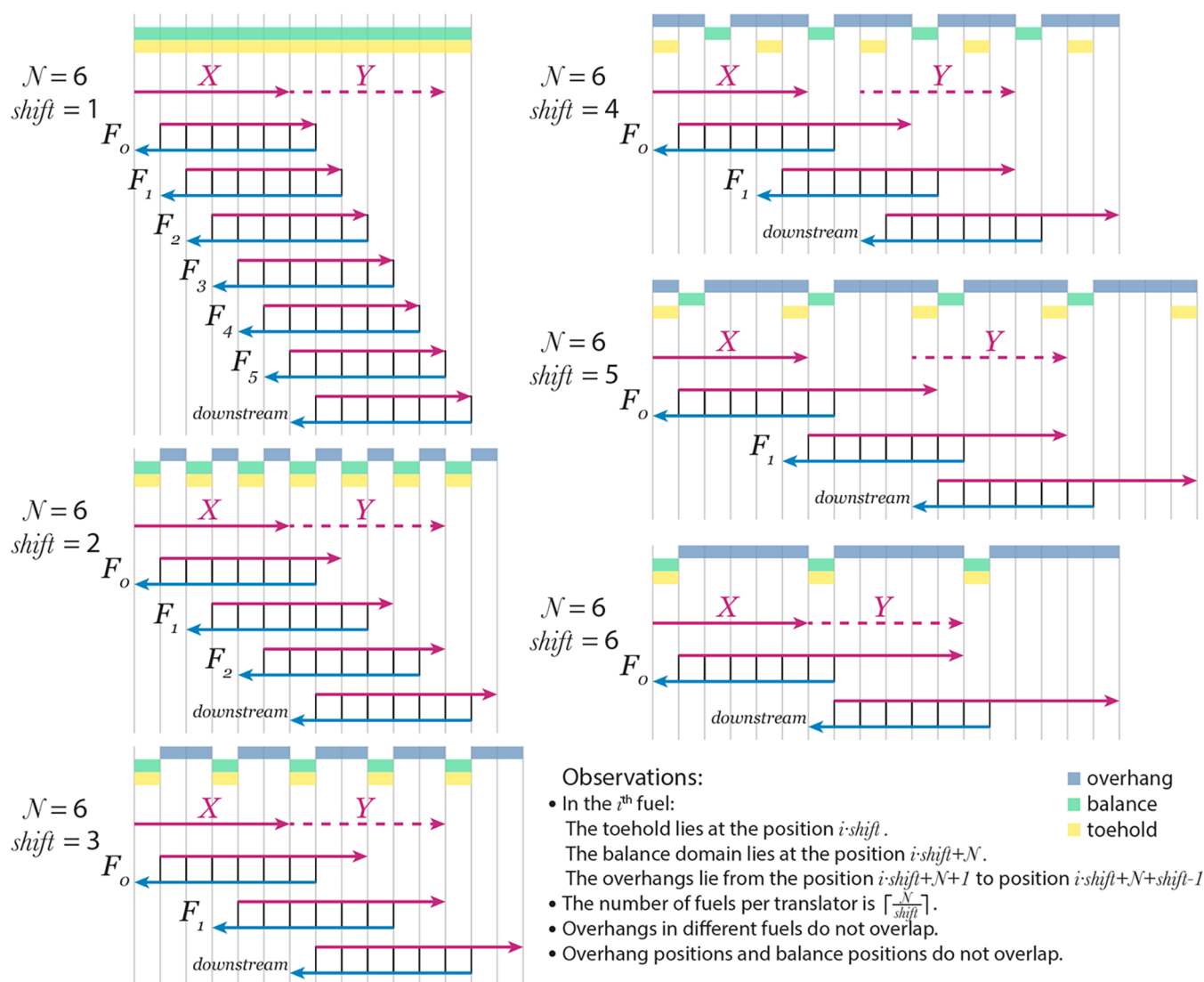


Figure 2. Initial configurations for linear enthalpy-neutral strand displacement cascades with $N = 6$ and all values of $shift$. For each combination of parameters, enough fuels are shown to output a signal of independent sequence from the input X (i.e., a full translator), as well as an additional (downstream) fuel. The dashed strand Y indicates the positions of the output signal strand that the downstream fuel takes as input (output region). The positions are colored on top of each design according to their position types (see legend) assuming that the cascade is extended both upstream and downstream. Top strands are colored in purple and bottom strands are colored in blue. Note that although drawn as a single copy, we consider a regime with multiple copies of each fuel. The observations are justified and explained in Section S3.

expected to have significantly less leak under the same conditions (see Section S8).

In an enthalpy-neutral cascade, every reaction is reversible, and thus the signal tends to spread out across the cascade more than in designs driven enthalpically by the bonding of toeholds. Moreover, adjusting our design parameters to increase the thermodynamic penalty to leak increases the number of such reversible reactions. However, we show that such signal spreading does not limit the capability of enthalpy-neutral cascades to effectively propagate signal, proving that the system completion level does not decrease arbitrarily with the depth of the cascade. Combined with our results on the thermodynamic penalty to leak, the lower bound on the output implies that, in principle, the “signal-to-noise” ratio can be made arbitrarily high. Because leak is reduced even at high concentrations, complex *EN* cascades may take minutes to complete instead of hours as is current state of the art.³

We conclude with experimental demonstrations of *EN* designs with different sets of parameters. The laboratory results confirm our theoretical categorization of designs with regard to leak and undesirable kinetic properties of toehold occlusion and spurious displacement, as well as the desired signal propagation. The leak reduction we observe compares favorably with the previously proposed *NLD* designs, in terms of the kinetic leak rate, the relative triggered signal to leak ratio and the maximum amount of leak generated. Despite the inevitable complexity of interactions between complementary regions in strand displacement cascades, our work suggests that this complexity can be effectively understood and programmed.

DESIGN SPACE OF ENTHALPY-NEUTRAL STRAND DISPLACEMENT CASCADES

System Description. In toehold exchange (Figure 1a), a DNA strand first binds the *toehold*—an unbound single-

stranded region (usually 5 to 8 nt) of the prehybridized complex. Subsequently, via a random walk process (3-way branch migration), this strand competes with the originally bound one, with complete displacement occurring after the dissociation of a symmetric toehold-size region. Single-stranded DNA fulfills the role of *signals* that carry information, while prehybridized DNA complexes (*fuels*) provide the material for the output signal.

We use *top strand/bottom strand* to indicate the strand at the top/bottom of a fuel complex respectively in our illustrations. Strands are divided into domains. Here all domains are assumed to be short enough that strands bound by a single domain can dissociate (see Figure 1a); such domains are typically said to be of toehold-length. Strands bound by two or more domains cannot dissociate. We assume that all domains are orthogonal (no cross-talk, see discussion in Section S9).

One or more single-stranded top domains on a fuel together are called an *overhang*. Single-stranded bottom domains are called *toeholds*. The domains at the right end of a double-stranded helix are called *balance domains*, since they break after strand displacement ensuring no net enthalpy change. (The balance domain can be thought of as either the “dissociation toehold” in toehold exchange reactions, or a long “clamp” in previous strand displacement designs used to “clamp-down” the ends of helices to reduce leak.) In our illustrations, domains aligned vertically have the same or complementary sequence, which is shown using *position* numbering as on top of Figure 1b. Note that as is typical of at-scale strand displacement experiments, there are many copies of each fuel although only one copy is drawn.

Linear Strand Displacement Cascades. When the input signal strand is present, it reacts with the first fuel displacing the top strand, which then triggers the downstream fuel (Figure 1c). A linear strand displacement cascade can be used to build *translators*, converting an input signal strand to an output signal strand whose *output region* is sequence independent of the input strand. By output region we mean the part of the strand that reacts with the downstream fuel. Figure 1c shows a translator consisting of two fuels (X and Y are sequence independent). We consider the composition of such translators into a cascade of arbitrary depth, where output region Y is the input region for the downstream translator.

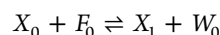
To design a linear, enthalpy-neutral strand displacement cascade, two parameters are necessary and sufficient: N ($N \geq 2$) represents the number of double-stranded domains in a fuel; *shift* represents the “distance” between consecutive single-stranded toeholds (see Figure 1b). Parameter *shift* should be between 1 and N to ensure signal can be transmitted to the next fuel. Figure 2 shows the diversity of designs when $N = 6$ for all valid values of *shift*.

Using 0 indexing, we assign domain types starting from the zeroth fuel which is responsible for the first strand displacement step (e.g., Figure 1c). The toehold of the zeroth fuel is located at position 0. A position can belong to one or more different position types: *toehold*, *overhang*, and *balance*, depending on which types of domains exist in that position. Each position type has its own color shown in the figures. (We assume the linear strand displacement cascade is extended both upstream and downstream, so that every bottom strand has exactly the same layout of position types on all the domains.) Basic properties and observations for the enthalpy-neutral strand displacement cascade are shown in Figure 2 and listed in Section S3.

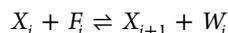
THEORETICAL RESULTS

Asymptotic Completion Level of Translator Cascades. With a cascade of irreversible or highly favored reactions, most of the input signal should propagate through to the end. However, enthalpy-neutral cascades are reversible and the amount of output decreases with the length of the cascade since the signal “spreads out” across the layers. Nonetheless, we argue that long cascades are feasible by proving a lower-bound on the amount of final signal output, such that this lower-bound is *independent of the length of the cascade*.

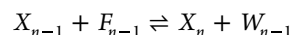
We simplify each enthalpy-neutral strand displacement reaction to be a bimolecular reversible reaction $X + F \rightleftharpoons Y + W$, where X is the input signal, F is the fuel, Y is the displaced strand of this reaction, and W is the waste species. The enthalpy neutral assumption implies that the equilibrium constant of each reaction can be treated as 1 (Section S4). Now suppose we have an n -layer reaction system starting with fuel (F_i , $i = 0, 1, \dots, n - 1$) concentrations at 1 and input X_0 at α :



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Theorem 1. Given a linear enthalpy-neutral strand displacement cascade with n layers, with fuel concentration 1 and input concentration α , in the limit $n \rightarrow \infty$ the equilibrium concentration of the output signal is $\alpha(1 - \alpha)$ when $\alpha \leq 1/2$, and $1/4$ when $\alpha > 1/2$.

The proof for this theorem is in Section S4.1. Thus, even in the limit of long reaction cascades, the amount of output is bounded from below. Note that the amount of output monotonically decreases with the number of layers (Section S4.1 and Figure S1); therefore, the theorem provides a lower bound on the amount of output produced by linear cascades of enthalpy-neutral reactions.

Kinetic Properties. In this section, we show that some designs are susceptible to undesired properties called *toehold occlusion* and *spurious strand displacement* to varying degrees depending on parametrization of N and *shift*. Both toehold occlusion and spurious strand displacement have the potential to slow down the kinetics of producing the output when the input is given due to the sequestration of fuels and/or signal species.

In the consideration of which undesired kinetic reactions to consider, we are guided by the rule of thumb that the input concentration is relatively low compared with the fuel concentration. Indeed, for our experimental regime, the fuels are always present at higher concentration than either the signal strands or (consequently) the waste products. Thus, for example, we do not consider spurious interactions between waste and signal strands.

Many of our arguments rely on showing whether regularly spaced positions with certain domains (such as toeholds) can intersect with other regularly spaced positions or intervals with different domains (such as overhangs). Those arguments are based on some useful lemmas in Section S2.

Toehold Occlusion. Stronger toeholds enable faster initiation of strand displacement;¹¹ however, if toeholds are too strong, besides slower dissociation required for toehold exchange reactions (i.e., dissociation of balance domains),

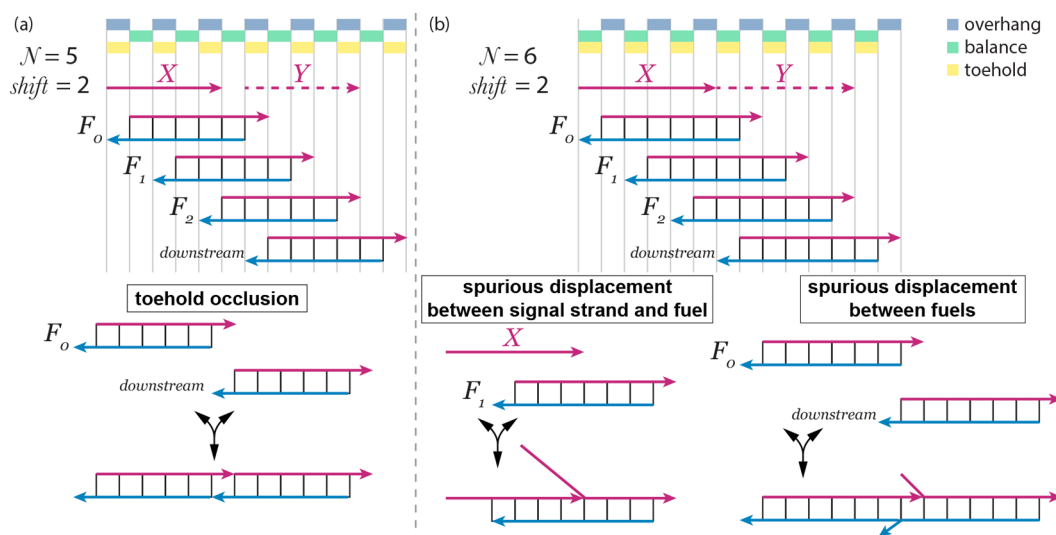


Figure 3. Examples of the properties of linear enthalpy-neutral strand displacement cascades. (a) Toehold occlusion in the design $N = 5$, $\text{shift} = 2$. (b) Spurious displacement in the design $N = 6$, $\text{shift} = 2$.

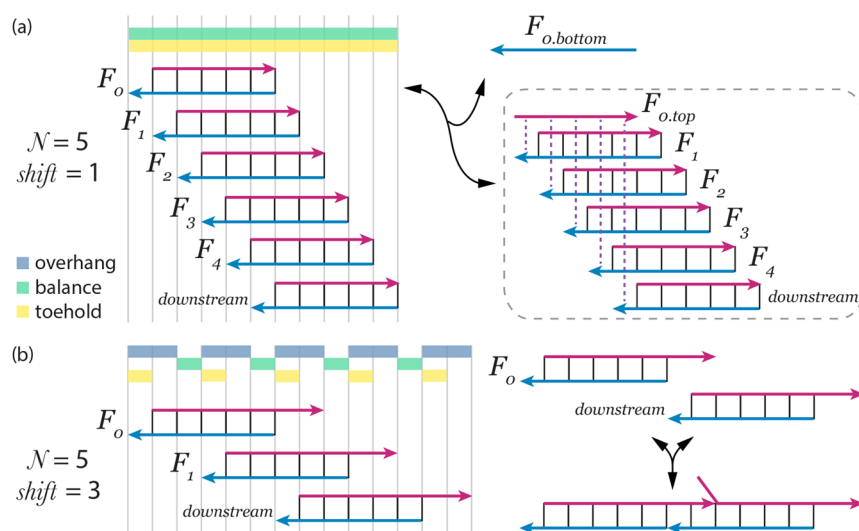


Figure 4. Examples of configurations reachable with spurious strand displacement. (a) Spurious strand displacement results in a configuration that requires a bimolecular reaction to undo. In the absence of input signal, multiple spurious displacement events could result in all of the bottom domains of one fuel being displaced by toeholds of other fuels. This results in a free (unbound) bottom strand. Although the multiple toehold displacement events to cause this reconfiguration are unlikely, once formed, it requires a slow bimolecular reaction to undo. The dashed gray box indicates the strands in the box are in one complex. The dashed purple lines indicate bound domains. (b) Spurious strand displacement results in a configuration that requires a unimolecular reaction to undo.

reaction kinetics also slows down due to undesired toehold binding.^{3,7} Since typically fuels are at high concentration, this is especially problematic when overhangs bind to toeholds of other fuels (when overhang and toehold positions overlap, Figure 3a). This creates *toehold occlusion*, which can significantly slow down the intended reaction. We say that toehold occlusion exists when there are overlapping overhang and toehold positions in different fuels (fuel–fuel toehold occlusion, Figure 3a). The following theorem (proof in Section S5.1) characterizes exactly the designs that avoid toehold occlusion. This characterization of toehold occlusion between fuels is also important for our thermodynamic analysis of leak in subsequent sections.

Theorem 2. *Toehold occlusion is not possible in a linear enthalpy-neutral strand displacement cascades if and only if N is a multiple of shift .*

Note that besides fuel–fuel toehold occlusion, signal strands can also bind the toehold of a fuel that is not their intended target (i.e., signal–fuel spurious binding). Spurious binding between signal strands and fuels can be avoided exactly when $\text{shift} = N$ (Lemma S5.2). We expect that such binding is less problematic for our systems than toehold occlusion because (1) signal strands are present in lower concentration than fuels, and (2) spurious binding is reversible and it does not prevent the signal strand from initiating intended displacement. While we do not further consider such signal–fuel spurious binding, we do consider potentially more problematic spurious

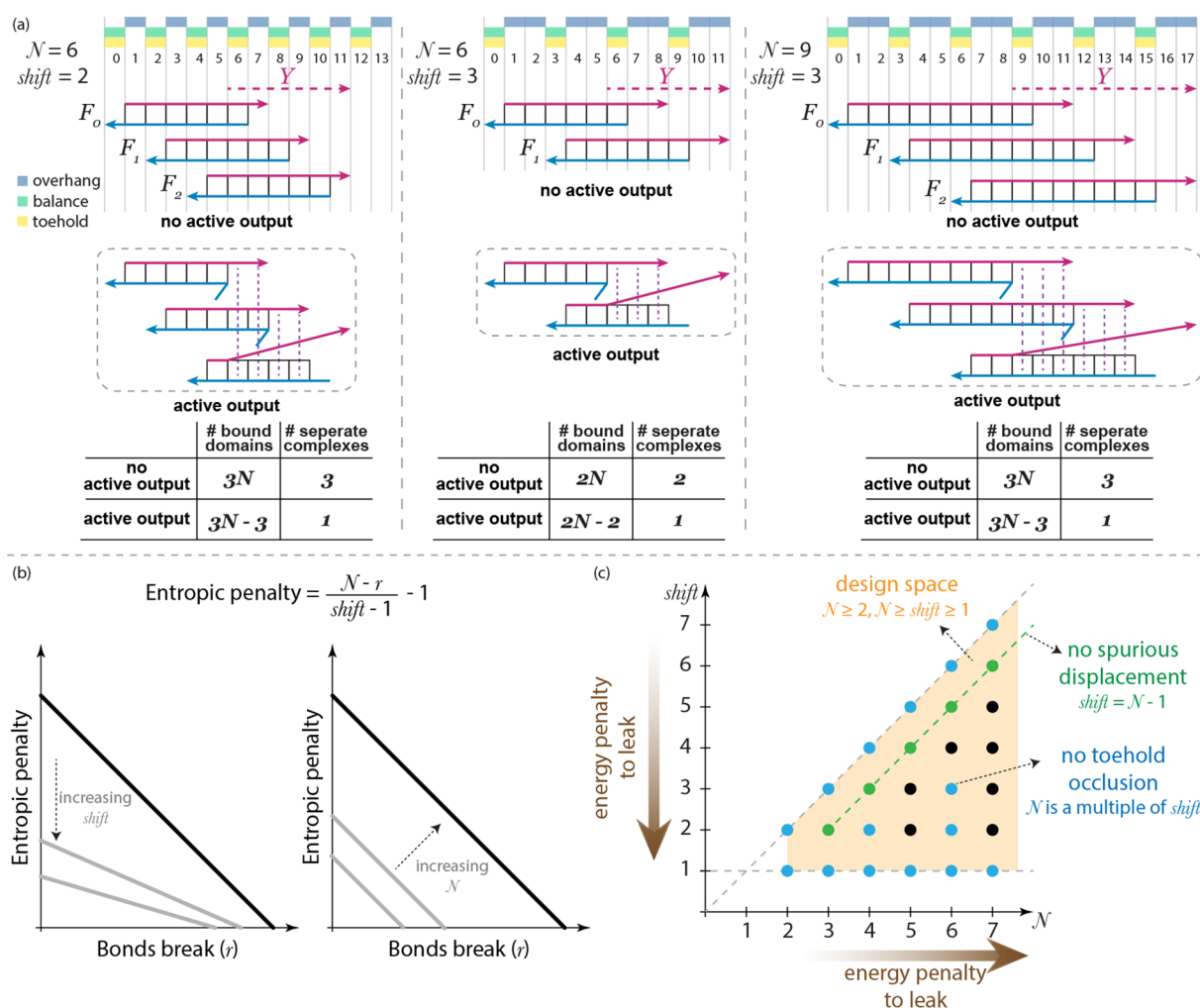


Figure 5. Thermodynamic and kinetic properties of linear enthalpy-neutral strand displacement cascades with different design parameters. (a) Example configurations showing the energy penalties for active output in three different designs. The dashed gray box indicates the strands in the box are in one complex. The dashed purple lines indicate bound domains. To have active output, positions 6, 8, and 10 in the design $N = 6$, $shift = 2$, positions 6 and 9 in the design $N = 6$, $shift = 3$, and positions 9, 12, and 15 in the design $N = 9$, $shift = 3$ need to break bonds. The shown active output configurations incur both an enthalpy (fewer bound domains) and entropy (fewer separate complex) penalty and are thus thermodynamically unfavorable. (b) The trade-offs between entropic and enthalpic penalty to leak. (c) The design space for linear enthalpy-neutral strand displacement cascades and their corresponding desired and undesired properties.

displacement (which requires at least 3-way branch migration to reverse) between signal strands and fuels in the next section.

Spurious Strand Displacement. Informally, spurious strand displacement includes any strand displacement reaction that is not in the desired reaction pathway. We include “partial” displacement in which a part of a strand is displaced but the strand remains attached. Spurious displacement could be initiated at a toehold or be “toeless”, enabled by spontaneous fraying at the ends of helices.²¹ The concern of this section is not with leak which results from spurious displacement when the input is not present (leak is covered by the thermodynamic analysis in a later section), but rather with the slowing of the intended triggering kinetics when the input is present by sequestering the species involved. Roughly speaking, our analysis here is kinetic: we count the points of initiation of spurious displacement. The more points of initiation of spurious strand displacement there are, the more sequestration of fuels and/or signal strands we expect, slowing down the kinetics of desired triggering when the input is present.

We consider spurious displacement both between fuels and also between fuels and signal strands. In the second type of spurious displacement, signal strands can be involved in numerous unproductive reactions, thus slowing (possibly significantly) signal propagation through every layer of the cascade—this is especially of concern when the input concentration is significantly lower than fuel. Note that we only characterize the point of initiation of spurious displacement with respect to the initial configurations. After initiation, more complex spurious displacement reactions involving intermediate products can occur such as shown in Figure 4a and Figure 8.

To define fuel–fuel spurious displacement, let the *left flank* be the domain that is immediately to the right of a toehold domain. We say that fuels F_i and F_j can undergo *spurious displacement initiated at position p* if p is the left flank or the balance domain of F_j and p is the toehold or overhang of F_i . In order to define signal–fuel spurious displacement, let the *i -th output region* (for any $i > 0$) consist of N positions $i \cdot \text{shift}$ to $i \cdot$

$shift + N - 1$. These positions in the signal strand are involved in the intended reaction with its downstream fuel, and sequestering these domains can slow down the desired reaction pathway. Signal strand S_i and fuel F_j ($j \notin \{i - 1, i\}$) can undergo *spurious displacement initiated at position p* if the output region of S_i contains p and p is the left flank or the balance domain of F_j . Both examples are shown in Figure 3b. In Section S5.3, we count the number of spurious displacement initiation events that a single fuel or signal strand can be involved in.

The following theorem (proof in Section S5.3) characterizes when both kinds of spurious displacement are avoided. Lemmas in Sections S5.3.1 and S5.3.2 count the points of initiation of spurious displacement between fuels, and between signal strands and fuels, respectively.

Theorem 3. *Spurious displacement is not possible in a linear enthalpy-neutral strand displacement cascade if and only if $N > 2$ and $shift = N - 1$.*

Of course, not all spurious displacement is equally problematic. Spurious displacement initiated with toehold binding (Figure 3b, left) is faster than toeless spurious displacement (Figure 3b, right). Spurious displacement of the same kind of species—e.g., fuel–fuel—can also have different effects. For example, Figure 4 shows contrasting examples with hard-to-reverse spurious reconfiguration (requiring bimolecular steps to undo) and those in which any spurious reconfiguration can be quickly undone (unimolecular steps are sufficient). To the first approximation, bimolecular reactions between spuriously produced species, which are in relatively low concentration, are expected to be substantially slower than unimolecular reactions. We prove that in certain cases such hard-to-reverse displacement is thermodynamically penalized (see Section S7); however, obtaining a more general characterization remains an area for further research.

Thermodynamic Properties: Thermodynamic Penalty to Leak. Although strand displacement systems are designed to not generate output assuming displacement only follows toehold binding, leak can result from spurious displacement reactions discussed above, or from more wholesale rearrangement of strands between complexes. While nonenthalpy neutral strand displacement schemes are typically metastable and leaky if left without input for a long time, we argue that our EN design has a programmable *thermodynamic penalty* to leak (comparison details in Section S8). Note that our thermodynamic analysis in this section contrasts sharply with the kinetic analysis of the previous section on spurious displacement. Rather than counting particular initiation points of spurious displacement, our arguments implicitly enumerate over all possible pathways by considering all possible configurations.

To study the thermodynamics of leak, we need to answer two questions about the system in the absence of input: What configuration is thermodynamically preferred and what penalty must be incurred to leak? Our thermodynamic arguments are based on “enthalpy”—counting the number of bonds formed (more bonds are more favorable) and “entropy”—the number of separate complexes (more complexes are more favorable).^{22–24} We acknowledge that our use of “enthalpy” and “entropy” are merely shorthand; there are entropy contributions of bond formation, and there are other components of entropy such as conformational entropy and the distinguishability of molecules. An example of analyzing the energy penalty to leak is shown in Figure 5a.

Varying design parameters N and $shift$ changes the thermodynamic penalty to leak. All of our combinatorial arguments are valid both in the single-molecule regime or when the fuels have multiple copies. Different fuels may also be present in different amounts.

Recall that the i -th output region (for any $i > 0$) consists of N positions $i \cdot shift$ to $i \cdot shift + N - 1$. A *configuration* is a matching between corresponding top and bottom domains, where each match is one bond. A configuration has an *active output* if some complex has open top domains in all N positions of some output region. These positions represent the domains in a top strand that can displace a strand of the downstream fuel. In the *initial configuration* each fuel complex is separate and unreacted (e.g., the configurations shown in Figure 2).

We first show that in the absence of input, the initial configurations are the thermodynamically preferred configurations for designs without toehold occlusion. Thus, the thermodynamic penalty to produce an active output can be properly based on the comparison between the active-output configuration and the initial configuration. We then discuss the enthalpic and entropic penalties, and the trade-offs between them. An active output has N unbound domains. At some of these positions, to produce an active output in the absence of input, it is sufficient to bring some complexes together without decreasing the overall number of bonds. This decrease in the number of separate complexes results in an entropic penalty. This entropic penalty can be traded off with an enthalpic penalty of breaking the corresponding bonds. However, at other positions, no matter how many complexes are brought together, producing an active output requires decreasing the number of bonds, which results in an unavoidable enthalpic penalty.

Note that the arguments in this section only apply to designs without toehold occlusion, because for these designs, in the absence of input, the thermodynamically preferred configurations are the initial configurations. However, for designs with toehold occlusion, depending on the parameters of the system (strength of toehold, concentration, temperature, etc.), the thermodynamically preferred configuration is variable and toeholds can be either occluded or not. (If occluding toeholds is preferred for a given reaction condition, the thermodynamically preferred configuration has the maximum number of possible bonds and the results about the enthalpic penalty still hold.)

Initial Configuration Is Thermodynamically Preferred without Toehold Occlusion. We first show that for designs without toehold occlusion, in the absence of input, the initial configuration is thermodynamically preferred. There might be trade-offs between enthalpy (number of bonds) and entropy (number of separate complexes), but if we find a configuration that simultaneously maximizes both, it is thermodynamically preferred. In our case, we will show that the initial configuration simultaneously maximizes both unless we are willing to lose a lot (N) of bonds. Thus, it is the thermodynamically preferred configuration in a wide range of experimental regimes.

In our proof of the following theorem, we gradually restrict the possible complexes of the thermodynamically preferred configuration through Lemma S6.2, Lemma S6.3, and finally by showing that they must consist of exactly one top and one bottom strand (Lemma S6.4). Combined with an analysis of the trade-off between bonds and separate complexes later

developed in Theorem 6, this yields the following result (proof in Section S6.1):

Theorem 4. *For designs without toehold occlusion, given a linear enthalpy-neutral strand displacement cascade of arbitrary depth, in the absence of input, we have the following: (i) Among the configurations with maximum bonding, the initial configuration is the unique configuration that maximizes the number of separate complexes. (ii) The initial configuration has at least N more bonds than any configuration with more separate complexes.*

Enthalpic Penalty to Leak. Our arguments about the enthalpic penalty are based solely on counting the difference in bonds between the thermodynamically preferred configuration and the configuration with active output. We show that the enthalpic penalty to leak can be increased arbitrarily by tuning the parameters N and $shift$.

Consider a position where, in the system as a whole, there are at least as many bottom domains as top domain. For this position to be single-stranded, we must lose one bond relative to the fully bonded configuration. Thus, it is sufficient to count the number of positions among the output region that do not have excess top domains (top-limiting positions, including toehold and balance positions; Observation S6.1, Lemma S6.6). For example in Figure 5a, positions 6, 8, and 10 in the design $N = 6$, $shift = 2$, positions 6 and 9 in the design $N = 6$, $shift = 3$, and positions 9, 12, and 15 in the design $N = 9$, $shift = 3$ are the top-limiting positions in the output region.

Theorem 5. *Given a linear enthalpy-neutral strand displacement cascade of arbitrary depth, in the absence of input, any configuration having active output has at least $\left\lceil \frac{N}{shift} \right\rceil$ fewer bonds than the maximum-bond configuration.*

The proof for this Theorem is in Section S6.2. This theorem implies that in the absence of input there is an enthalpic penalty of $\left\lceil \frac{N}{shift} \right\rceil$ bonds to generate an active output from a maximum-bond configuration. Note that in the designs without toehold occlusion, the initial configuration maximizes the number of bonds and so can act as the unique point of comparison (via Theorem 4). The theorem still applies to designs with toehold occlusion but the thermodynamically preferred configurations are not necessarily maximum-bonded so the relevance of the point of comparison is not clear.

In contrast, when the input is present, the signal can be propagated all the way with no net enthalpic cost. Thus, by increasing N we can enlarge the enthalpic penalty to leak without increasing the enthalpic penalty to correct output. The enthalpic penalty holds even at high concentrations when there is little entropic penalty of joining two complexes into one, and thus sets the lower bound on the required energy for leak independent of concentration.

The enthalpic penalty shows that leak results in fewer bonds compared with maximum bonding. Moreover, Theorem 4(ii) shows that leak cannot compensate for the loss of bonds with the thermodynamic benefit of creating additional separate complexes (unless N bonds break). Indeed, in the next section we show that leak results in a decrease in the number of separate complexes, and that this entropic penalty can be made arbitrarily large by varying N and $shift$.

Entropic Penalty to Leak. In this section we show that no matter how strands are rearranged in our enthalpy-neutral strand displacement cascades, any configuration with active output has fewer separate complexes—thus it incurs an entropic penalty. As the point of comparison, we count the

difference in the number of separate complexes between the thermodynamically preferred configuration as captured by Theorem 4 and the active-output configuration.

By Theorem 5, we know that the enthalpic penalty has a lower bound of $\left\lceil \frac{N}{shift} \right\rceil$. Thus, the arguments below assume that the configuration with active output has r bonds fewer than maximum, where $r \geq \left\lceil \frac{N}{shift} \right\rceil$. Beyond this lower-bound on r , the entropic penalty can be traded-off for an decreased enthalpic penalty by increasing r . Intuitively, forcing fewer bonds to form gives more flexibility to produce active output without bringing as many complexes together.

To count the difference in the number of separate complexes, we can equivalently count the number of top strands associated in a complex in the active-output configuration (since the thermodynamically preferred configuration has only one top strand in every complex by Lemma S6.4). Then by a combinatorial argument based on counting the types of positions in top and bottom strands (Lemma S6.9, Lemma S6.10, Lemma S6.11, and Lemma S6.12), we show that for active output there must exist some complex with at least $\frac{N-r}{shift-1}$ top strands (Lemma S6.11 and Lemma S6.12), implying the following theorem:

Theorem 6. *For designs with $shift \neq 1$ and no toehold occlusion, given a linear enthalpy-neutral strand displacement cascade of arbitrary depth, in the absence of input, any configuration with active output and r ($r \in \mathbb{N}$, $r \geq N/shift$) bonds away from maximum has at least $\frac{N-r}{shift-1} - 1$ fewer separate complexes than the initial configuration.*

The proof for this Theorem is in Section S6.3. This theorem implies that in the absence of input there is an entropic penalty of $\frac{N-r}{shift-1} - 1$ units to produce an active output. In the case of $shift = 1$, active output requires an enthalpic penalty of $r = N$, but can result in one additional separate complex. Figure 5b visualizes the relationship between N , $shift$, and r captured by the above theorem.

As in the case of enthalpy, when the input is present, the signal can be propagated all the way with no net entropic cost. Combined with the results of the previous sections, if we fix $shift$ and increase N , both the enthalpic (Theorem 5) and entropic penalty (Theorem 6) to leak increases, while the desired triggering signal reaches an asymptote (Theorem 1).

Trade-Offs between the Properties. We have shown that different choices of N and $shift$ yield EN designs with varying thermodynamic and kinetic properties. We desire a large thermodynamic penalty to leak, which requires a large ratio $N/shift$ (Theorems 5 and 6). However, only designs with $shift = N - 1$ can avoid spurious strand displacement (Theorem 3), which is incompatible with a large $N/shift$ ratio. In fact, designs with the largest thermodynamic penalty to leak also have the most potential spurious interactions between fuels and signal strands (Lemma S5.7). (Compare the spurious interactions of the X input signal with fuels other than its intended target for different $shift$ in Figure 2.)

Besides spurious displacement, the other undesired kinetic property is toehold occlusion. While the absence of toehold occlusion is compatible with a large thermodynamic penalty (Theorem 2), there is a fundamental incompatibility between the two kinetic properties:

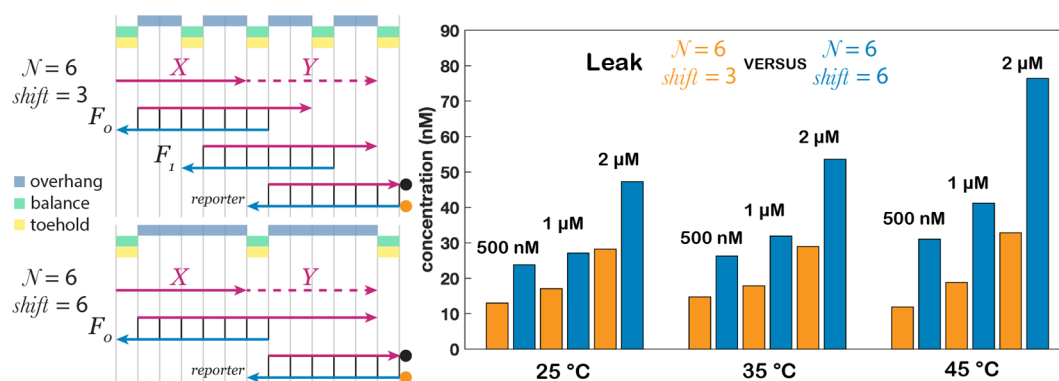


Figure 6. Comparison of designs with different energy penalties to leak. Signals at different thermodynamic equilibrium (25 °C, 35 °C, and 45 °C) after a slow annealing from 90 °C (0.06 °C/min) without input strand. The design with parameter set $N=6$, $shift=3$ has less relative leak. The top strand of the reporter is labeled with a quencher (black circle). The bottom strand of the reporter is labeled with a fluorophore (orange circle). The concentrations for fuels and reporters are labeled in the figure.

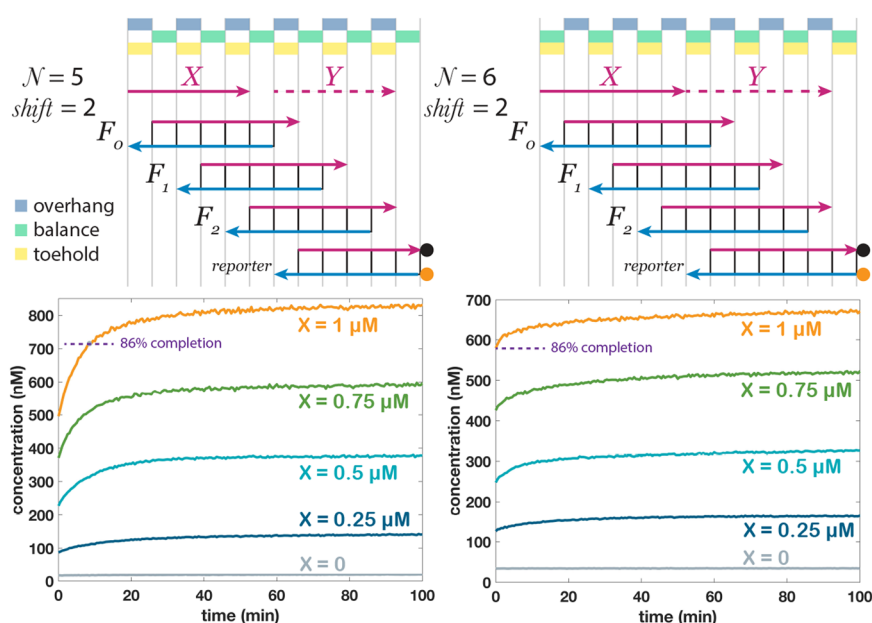


Figure 7. Comparison of kinetics of designs with ($N=5$, $shift=2$) and without ($N=6$, $shift=2$) toehold occlusion. We define completion as the signal level at 100 min. The design with the parameter set $N=6$, $shift=2$ is significantly faster. The concentrations of reporter and fuels were 2 μM. The reaction temperature was 25 °C. As in all kinetic curves in this work, time 0 represents the first data point measured, which is about 2 min after samples were mixed.

Corollary 1. *There is no design that avoids both toehold occlusion and spurious displacement.*

In summary, there is no best design with respect to all thermodynamic and kinetic properties studied here. Instead, the entire taxonomy we develop informs the choice of translator design, and one should be chosen based on the expected conditions of its planned use. For example, high concentration conditions may be best served by a design with no toehold occlusion and a balance between its enthalpic penalty to leak and its potential number of spurious displacement reactions. Figure 5c summarizes the kinetic and thermodynamic results of the previous sections.

EXPERIMENTAL RESULTS

In this section, we experimentally investigate the theoretical results we developed in prior sections. First we compare two sets of parameters of N and $shift$, and observe that the one that

is predicted to have a larger penalty to leak is indeed less leaky. Then we compare designs with and without toehold occlusion, obtaining results consistent with slower kinetics when toehold occlusion is expected. In addition, we study one spurious displacement pathway, supporting the relevance of the theoretical results about avoiding the initiation of spurious displacement. We also experimentally compare leak in enthalpy-neutral strand displacement cascades with “leakless” schemes from prior work.^{8,9} We conclude by testing desired triggering and leak in longer cascades, showing small overall leak but robust desired triggering in accord with the theory.

Designs with Different Energy Penalties to Leak. We experimentally tested the energy penalties to leak among two sets of design parameters (Figure 6). The $N=6$, $shift=3$ design has 2 units of enthalpic cost and 1 unit of entropic cost to leak. The $N=6$, $shift=6$ design has only 1 unit of enthalpic cost, and no entropic cost. We used fluorophore and quencher

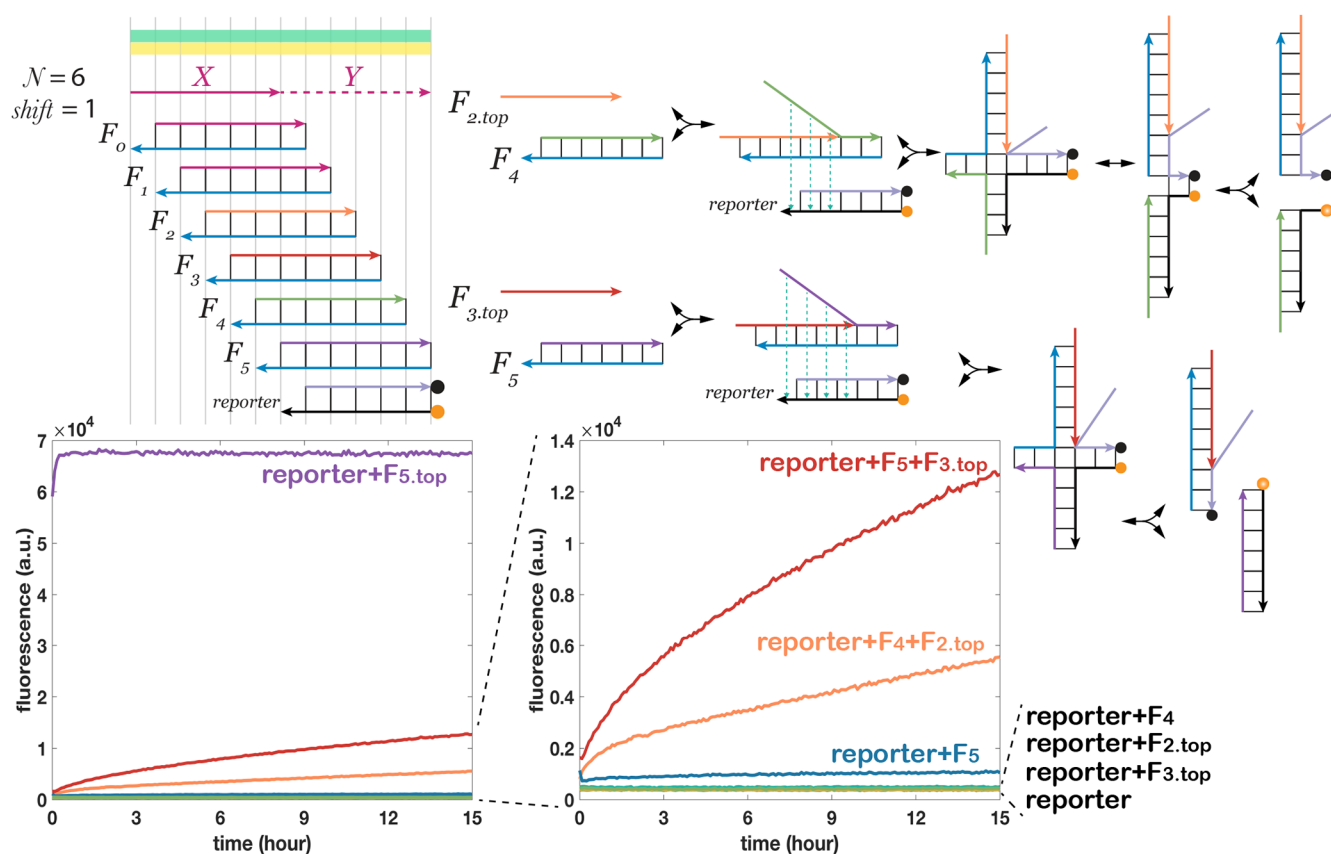


Figure 8. Spurious displacement pathways and kinetics of spurious interactions at the design $N = 6$, $shift = 1$. The concentrations of reporter and fuels were 500 nM. The input at each layer was 100 nM. The reaction temperature was 25 °C.

labeled “reporter” complex to detect output signal. The separation of the two strands of the reporter results in a measured fluorescent signal. (See Section S11 for details of normalization.)

Experiments confirm that the $N = 6$, $shift = 3$ design has less leak than the $N = 6$, $shift = 6$ design (Figure 6). The leak concentration of the $N = 6$, $shift = 6$ design increased with temperature as is consistent with an enthalpic penalty. In contrast, the effect of entropy on the equilibrium should be independent of temperature. Leak in the $N = 6$, $shift = 3$ design did not have significant temperature dependence, which may be due to the entropic contribution to the leak penalty.

Since the energy penalty to leak of the $N = 6$, $shift = 6$ design is not entropic, the leak concentration is expected to be proportional to the initial concentration of the reacting species. However, the experimental leak concentration does not double when the initial concentration doubles. This deviation could be caused by some reactions changing entropy that are not captured in our model, such as sequence-dependent spurious binding or the fluorophore-quencher interaction. (Note that this design is predicted to not have toehold occlusion.)

In the presence of input signal, the two designs produce similar amounts of output signals (Figure S4). (Indeed, the low-leak $N = 6$, $shift = 3$ design has slightly higher output; we do not directly account for this difference with our theoretical results.)

Designs with and without Toehold Occlusion. Our theoretical results classify designs with toehold occlusion (Theorem 2), which can hinder the kinetics of desired strand displacement. Here, we experimentally compare the designs

with ($N = 5$, $shift = 2$) and without ($N = 6$, $shift = 2$) toehold occlusion. As shown in Figure 7, at various concentrations of input, the design without toehold occlusion ($N = 6$, $shift = 2$) reaches completion faster than the design with toehold occlusion ($N = 5$, $shift = 2$). When the input concentration is 1 μ M, the design without toehold occlusion reaches 86% of completion before the first measured data point, but it takes 9 min for the design with toehold occlusion to reach the same completion level.

Note that these two designs share the same toeholds, but there are slight differences between the energies of the toehold and balance domains. These thermodynamic differences actually favor the design with toehold occlusion, and thus cannot account for the kinetic slowdown. (These differences likely account for the greater completion level of the design on the left.) Further, note that, in fact, both designs have spurious binding between signal strands and fuels (Lemma S5.2), and the design with toehold occlusion has fewer spurious displacement reactions, but is still slower than the design without toehold occlusion. This suggests that toehold occlusion has a stronger impact on the kinetics than spurious displacement in this case.

Spurious Displacement Pathway with Signal Strands.

Spurious displacement may result in the formation of complex multistranded structures or intermediates, which could interfere with desired kinetics. Theorem 3 classifies designs with and without spurious displacement overall, and Lemmas S5.6 and S5.7 separate the cases of spurious displacement with the input absent or present. To experimentally investigate the signal-fuel spurious displacement pathways, we chose the

design with $N = 6$, $shift = 1$, which has the maximum amount of signal-fuel spurious displacement among $N = 6$ designs (Lemma S5.6).

Figure 8 shows two possible spurious displacement pathways involved with signal strands and the fuels that are not the designed targets of the signal strands. The signal strand first binds and partially displaces the top strand of the fuel, exposing a single-stranded portion of the top strand. The single-stranded portion can then interact with other downstream species. We use a reporter to represent the downstream species and detect the reaction pathway. After a 3-way branch migration and then a 4-way branch migration process, the fluorophore and quencher in the reporter separate. The gradual increase of fluorescence only in the presence of both an upstream fuel and a signal that can spuriously react with the fuel was consistent with the described spurious displacement pathway.

Leak Reduction Compared to Designs with Only an Entropy Barrier. Prior work (NLD design) based leak reduction on a kinetic barrier wherein multiple separate complexes needed to colocalize to generate leak (entropy barrier).^{8,9} Our EN design has a stronger guarantee of correct behavior in two ways: (1) by substituting an overall thermodynamic penalty for a kinetic barrier, (2) by basing this penalty on both entropy and enthalpy. (See detailed analysis in Section S8.) To experimentally compare leak reduction of our EN design to the prior work, we chose one of the parameters $N = 5$ and $shift = 3$ with the desired property that leak requires at least two units of enthalpic penalty (breaking two bonds) compared with the maximum-bond configuration (Theorem 5). We compare with the typical leaky translator with no leak reduction (SLD design) and the leak-reducing NLD design with “redundancy” parameter 2 (DLD design)^{8,9} (Figure 9a).

For a most fair comparison, in our experiments, these three translator designs share the same sequence space and all the fuels have the same toeholds. Moreover, the DLD and the EN design share the same reporter complex. The detailed comparison for the designs at the sequence level is shown in Figure S9. Figure 9b compares the amount of leak at thermodynamic equilibrium after annealing. Annealing serves as a proxy for the upper bound on leak expected in an experiment as time goes to infinity. (Kinetic measurements of leak in the EN design are shown in the next figure.) The EN design was observed to have an order of magnitude less leak than the DLD design and about 2 orders of magnitude less than the SLD design. The fuel concentrations used here are 10–100 times larger than the typical concentrations used in strand displacement systems, showing that EN schemes can reduce leak in the fast (high-concentration) regimes.

Although enthalpy-neutral strand displacement is not driven by the formation of new bonds as are the SLD and DLD designs, we observed that the triggered signal is only around three times lower than full triggering (Figure S5). Thus, the signal-to-leak ratio of the EN design is 30 times higher than of the SLD design and four times higher than of the DLD design (Figure 9b inset).

Desired Triggering and Leak in Longer Cascades. Beyond a single translator, we wanted to know (1) how much leak (without input signal) and (2) desired output signal (with input signal) a translator cascade can generate as the number of translators increases.

Figure 10a shows that in the time period of the experiment, in the absence of input signal strand, the translator cascades of

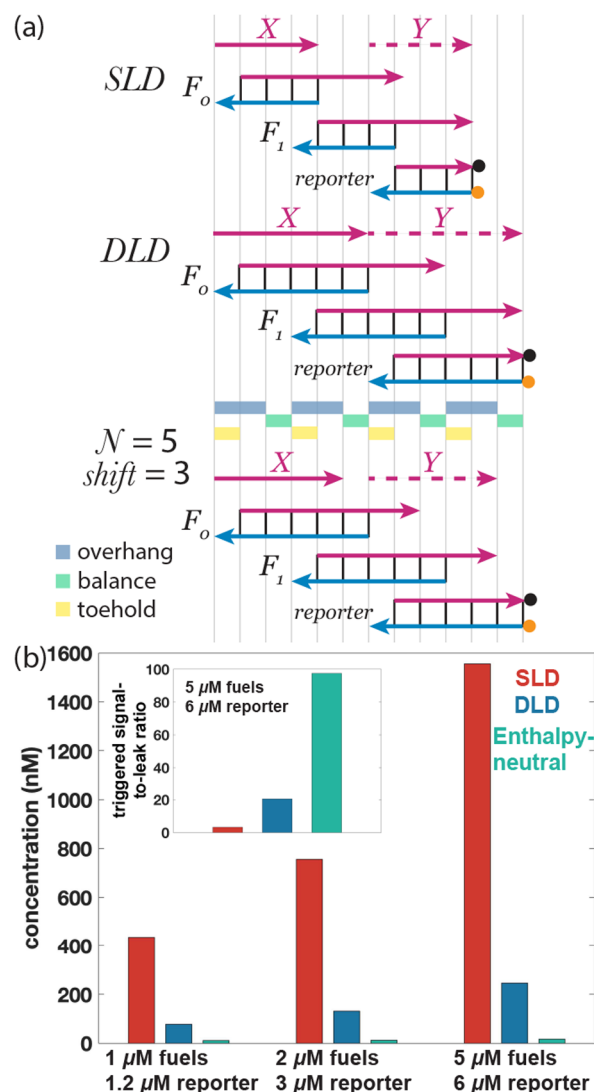


Figure 9. Comparison of leak between (a) the SLD, DLD, and EN design ($N = 5$, $shift = 3$) from the perspective of (b) the thermodynamic equilibrium. The experiments measuring the leak concentration at the equilibrium do not contain the input signal strands. Fuels and reporter were slowly annealed (0.06 °C/min) from 90 to 25 °C within 18 h. (Inset) The triggered signal-to-leak ratios of the NLD design and the EN design. The SLD and the DLD designs were assumed to reach full completion. The triggered signal for the EN design was shown in Figure S5. The concentrations for fuels and input were 5 μM, and the reporter 6 μM. The reaction temperature was 25 °C. For simplicity, the clamp domains in the SLD and DLD designs (which extend the bottom strand by 2 nt) are not shown here but are illustrated in Figure S9.

1 to 6 fuels (1 to 3 translators) all show no apparent leak. In the presence of input signal, the completion level decreases with the number of layers. However, as more layers are added, the completion level does not decrease linearly, and indeed seems to approach an asymptote—behavior consistent with our theoretical prediction. Note that Theorem 1 is not expected to give exact quantitative results for our system because it assumes all strand displacement reactions have $\Delta G = 0$, which is hard to achieve perfectly in experiments considering the energy differences between the toeholds and balance domains and fluorophore and quencher interactions.

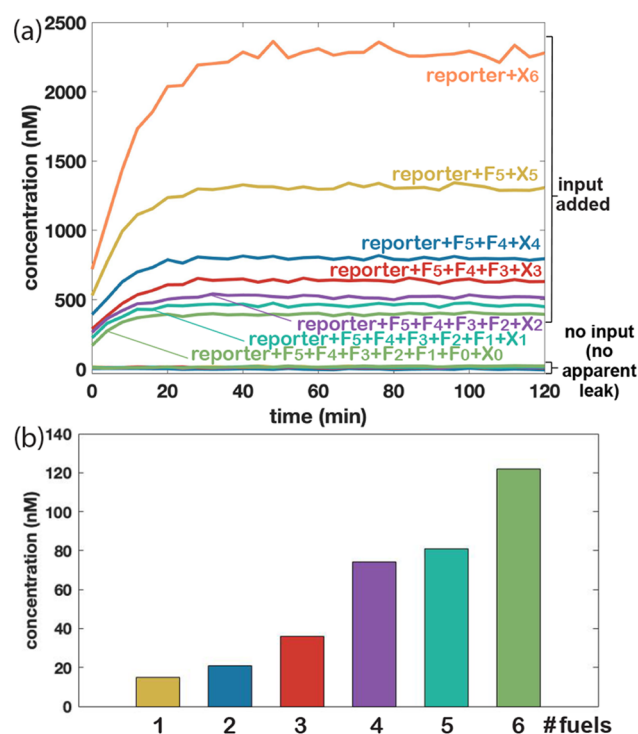


Figure 10. Kinetics and thermodynamic equilibrium of translator cascades with the *EN* design ($N = 5$, $shift = 3$). (a) Kinetic behavior in the presence and absence of input signal, for cascades of different length. (b) The total amount of leak in the absence of input signal at thermodynamic equilibrium measured after annealing, for cascades of different length. The concentrations of the reporter and the fuels were around $5 \mu\text{M}$. The concentration of each input was $2.5 \mu\text{M}$. The reaction temperature was 37°C . Note that X_i in the figure refers to the input to fuel F_i , and X_6 is the input to the reporter. (Note that the actual reporter used is one domain shorter than the $N = 5$ design and the overhand on F_5 was shortened to compensate (Figure S10), which is not expected to affect the completion level.).

Figure 10b studies the leak of translator cascades of varying depth at thermodynamic equilibrium. The leak at equilibrium increases as the number of fuels increases. These results are consistent with the expectation that leak increases arbitrarily with the length of the cascade if design parameters N and $shift$ are fixed; maintaining or increasing the signal-to-noise ratio requires increasing $N/shift$. Nonetheless, with these design parameters even if there are 6 fuels (3 translators), the total leak concentration is still less than 3% of the fuel concentration. Indeed, it is about 5 times less than the leak concentration for a single translator in the *DLD* design, and it is an order of magnitude less than that in the *SLD* design. Note that although the absolute amount of triggered signal of the *EN* design is lower than that of the *DLD* design, the relative triggered signal-to-leak ratio of the *EN* design is still higher than the *DLD* design. The results suggest that the *EN* design could be preferable, especially when absolutely smaller leak is required, such as when concatenating the translators with downstream catalytic or autocatalytic systems.

DISCUSSION

The design space of enthalpy-neutral strand displacement cascades shows a surprising range in terms of kinetic and thermodynamic behaviors which was previously not explored. The full diversity of enthalpy-neutral linear-topology strand

displacement cascades is accessible by varying two parameters N and $shift$, with different susceptibilities to toehold occlusion and spurious displacement, and rigorous guarantees on the enthalpic and entropic penalties to leak determined by these parameters. We formalized the analysis describing how the enthalpic and entropic penalties to leak can be raised arbitrarily, and captured the inherent trade-off between the two penalties. The enthalpic penalty argument is particularly germane for the high concentration regime where the entropic penalty for joining complexes is smaller. We further proved that certain parameter values result in other desirable properties like no spurious displacement and no toehold occlusion. Since no design satisfies all desired properties, understanding the taxonomy of designs is necessarily to tune the trade-offs and make proper design choices.

In addition to theoretical analysis (Section S8), we experimentally tested designs with different energy penalties to leak, confirming different degrees of leak reduction and improvement over the previous *NLD* design. Our experiments also corroborated slower kinetics due to toehold occlusion in otherwise similar designs. Finally, in identifying spurious displacement reaction pathways, the experiments confirm the rationale for studying spurious displacement.

The scale of current strand displacement systems exceeds the capabilities of thermodynamic modeling tools²⁵ and (even more so) of kinetic simulators.^{26–28} Thus, it is challenging to understand large-scale molecular behaviors due to the increasing complexity. However, combinatorial arguments like those espoused in this work apply to infinitely many, arbitrarily large systems. Our analysis is analogous to rigorous proofs of algorithm correctness in computer science, in which a single proof applies to all possible inputs that the algorithm may have. While this work focuses on the linear topology, we hope that desired and undesired behaviors in enthalpy-neutral systems with nonlinear topology—such as taking multiple inputs (fan-in) or producing multiple outputs (fan-out)—could be similarly analyzed in future work. More generally, we hope that such lines of reasoning could be developed for even broader classes of engineered molecular systems, helping to bridge the gap between electronic and molecular algorithms.

METHODS

Proofs. All the proofs are in the SI.

Sequence Design. The details for DNA sequence design principles are in the SI.

DNA Oligonucleotides. DNA oligos were synthesized by Integrated DNA Technologies (IDT). The unlabeled oligonucleotides were purchased PAGE purified by IDT. The oligonucleotides with fluorophore or quencher modifications were ordered HPLC (high-performance liquid chromatography) purified by IDT. Upon arrival, these DNA oligonucleotides were suspended in Milli-Q water. The concentration of each strand was quantified by NanoDrop. The absorbance at 260 nm was recorded and the concentration (c) was calculated as $c = [\text{Absorbance}]/\epsilon$, where ϵ is the extinction coefficient provided by IDT.

Fuel and Reporter Preparation. Fuels and reporters were prepared according to prior work.⁹ Fuels and reporters were annealed in TAE/ Mg^{2+} buffer (40 mM Tris, 20 mM acetic acid, 1 mM EDTA, and 12.5 mM Mg^{2+} , pH ≈ 8.0) with 10% excess of top strands from 90 to 25°C within 1.5 h. PAGE purification was then conducted to remove malformed structures or single-stranded DNA³ from fuels. Purified

duplexes were quantified by NanoDrop again and also concentrated by centrifugal filters (Amicon Ultra-0.5 mL, 10K device) to achieve a high stock concentration. Reporters were not purified. The extinction coefficients for complexes were estimated the same way as in prior work:⁹ adding up the estimated extinction coefficients for the single-stranded and double-stranded parts.

Fluorescence Measurement. Experiments for the enthalpy-neutral DNA strand displacement cascades with parameters $N = 5$, $shift = 3$ were measured by BioTek Cytation 5. All other experiments were measured by BioTek Synergy H1. The NBS (nonbinding surface) 384 well plates with clear flat bottom (Corning #3544) were used. Fluorescence was measured from the bottom. The excitation/emission wavelengths were set to 575/610 nm for the fluorophore ROX. For Cytation 5 measurement, the excitation bandwidth was fixed at 9 nm and the emission bandwidth was fixed at 20 nm.

For kinetic experiments, all the samples were mixed and prepared at room temperature (close to 25 °C), and then quickly transferred to a plate (within 2 min). Time 0 in plots represents the moment the first data point was measured.

For equilibrium experiments, all the samples were annealed from 90 °C to desired temperatures with 20 min per degree. The plate was incubated in the plate reader at the desired temperatures for 10 min prior to data collection to let the samples reach the same temperature as the experimental setting.

Data Fitting and Normalization. The details of data fitting and normalization are in the SI.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acssynbio.2c00356>.

Proofs, theoretical comparison between the EN and the NLD design, sequence design, data analysis, experimental data for triggered signals for some EN designs, and tables of DNA sequences (PDF)

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Author Contributions

B.W. initiated the project; B.W., C.T., and D.S. designed the research; B.W. and C.T. proved the theorems for kinetic properties; B.W. and D.S. proved the other theorems; B.W.

performed experiments and analyzed data; B.W., C.T., and D.S. wrote the paper.

Notes

The authors declare no competing financial interest.

[†]Preliminary versions of some of the results in this paper appeared in conference proceedings.²⁹

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